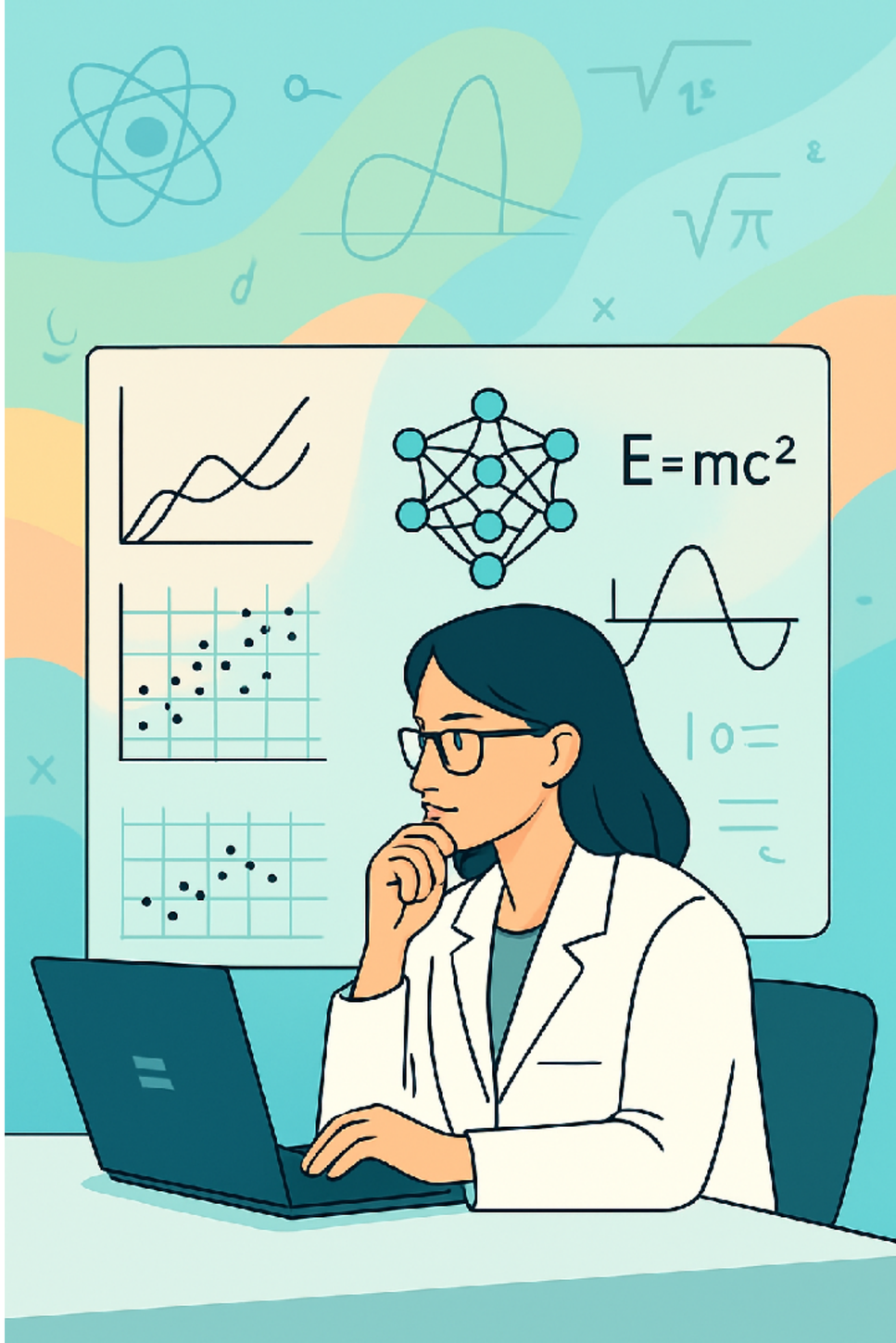
Statistics and Data Analysis

Week 1

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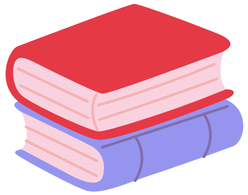
UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA



Course info

- Course of the master degree in physics — **6 ECTS (48 hours)**
- Thursday at 12:45—14:30 (room U2-04) and
  Friday at 13:45—15:30 (**room U2-01**)
- In person course (no live stream) with hands-on sessions
- Videos, slides and Jupyter notebooks will be availabe on elearning

Course material



Books, references and material

Statistics

Cowan - "Statistical Data Analysis"
Pruneau - "Data analysis techniques for physical scientists"
Metzger - "Statistical Methods in Data Analysis"

ML and data analysis

James - "An Introduction to Statistical Learning"
Prince - "Understanding Deep Learning"
Goodfellow - "Deep learning"
Mackay - "Information Theory, Inference and Learning Algorithms"
Fernandez-Granda - "Probability and statistics for data science"
Sivia - "Data Analysis: A Bayesian Tutorial"

Intro to Python

The following resources are meant to introduce basic elements of Python programming that will be useful during the course. The following Jupyter notebooks can be executed directly in the browser from Colab. Make sure you have installed the Google Colaboratory extension to your google drive.
Intro to Python
Intro to numpy
Intro to matplotlib
Intro to Scipy

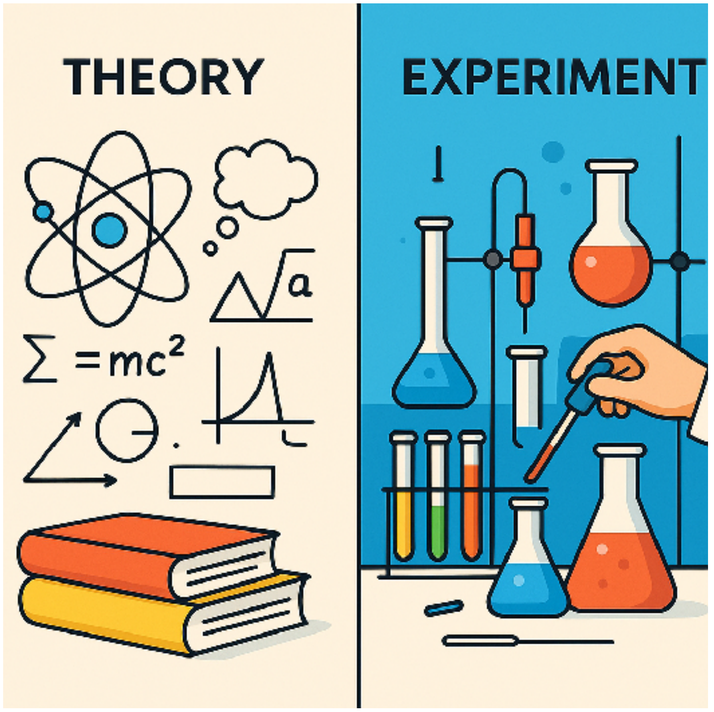
The 2nd year physics degree course "Laboratory of Statistics" provides excellent introductory lectures on python, data analysis libraries such as numpy, matplotlib and scipy.stats as well as basic probability and statistics techniques.

They can be found at this link:
UnimibFisicaLabStatPythonWebBook

What is statistics?

Theory ↔ Statistics ↔ Experiment

- Theory defines a model/hypothesis
- Experiments produce data with uncertainties
- Tasks of statistical data analysis:
  - Estimate unknown parameters and their uncertainties
  - Goodness-of-fit and hypothesis testing to assess model validity
  - Model comparison to enable discovery claims



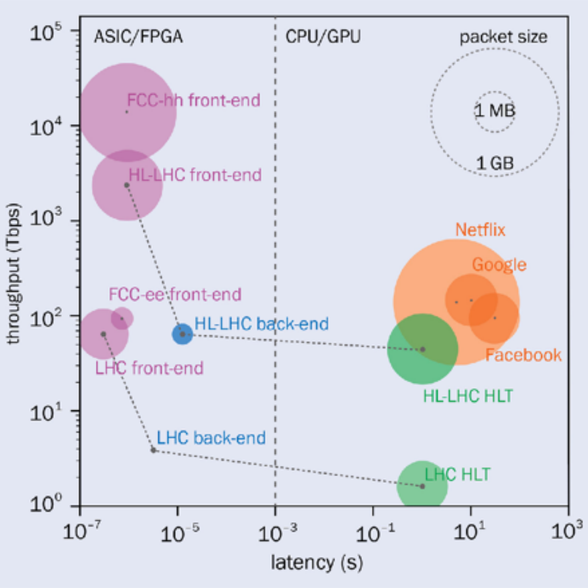
Statistical data analysis

- Extremely important in our data-rich world
  - Data-based decision making and uncertainty/risk assessment
  - Understanding complex problems (patterns, relationships, trends)
  - Improving efficiency and performance
  - Navigating media and information with a scientific mind
- *Modern physics also faces problems of huge data throughput*

ourworldindata.org

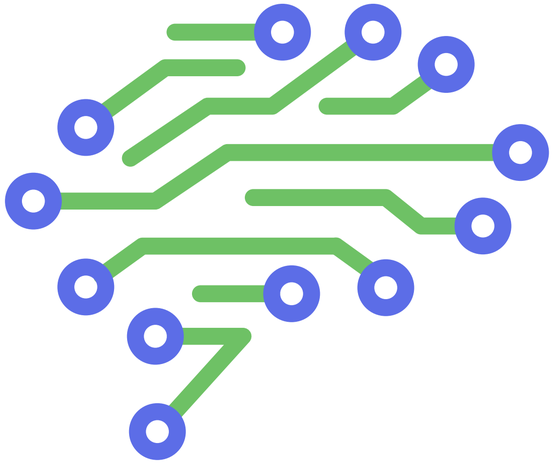


LHC data throughput



Machine Learning

- Machine Learning is statistical inference at scale
- Model training is parameter estimation
  (Loss function $\simeq$ Likelihood)
- Uncertainty quantification matters in all
  applications of ML (not only in physics)
- Evaluation of ML requires stat metrics,
  hypothesis testing, model comparison



Program of the course

- Week 10 — Probability Theory and Monte Carlo Foundations
- Week 11 — Frequentist Inference: Hypothesis Testing and Maximum Likelihood
- Week 12 — Precision Estimation and Confidence Intervals
- Week 13 — Bayesian Inference and Model Comparison
- Week 14 — **No classes (Easter holidays)**
- Week 15 — Statistical Learning and Regularization
- Week 16 — Foundations of Deep Learning and Unsupervised Methods
- Week 17 — Neural Network Optimization and Training Strategies
- Week 18 — **No classes (May 1st)**
- Week 19 — Structured Neural Architectures and ML Uncertainty
- Week 20 — Markov Chain Monte Carlo Methods
- Week 21 — Unfolding and Statistical Inverse Problems
- Week 22 — End-to-End Analysis and Integration
- Week 23 — Buffer for final project
- **Exam**: oral exam on submitted final project and questions on the topics of the course

= includes Machine Learning

Program of the first 4 weeks (March)

Days	Thu 12:45 - 14:30 (2h)	Fri 13:45 - 15:30 (2h)
05/03 06/03	- Probability: foundations and interpretations - Practice with PDFs, CDFs, Quantiles - Joint PDFs, moments, correlation, transformations	- Practical CLT and the lognormal - Linear error propagation - MC methods - **Exercise**
12/03 13/03	- Frequentist hypothesis testing - Optimal tests, Neyman-Pearson, p-value - Estimators and properties	- Likelihood - MLE - **Exercise**
19/03 20/03	- Asymptotic MLE properties and MVB - Advanced MLE applications and MC methods - Frequentist Confidence intervals	- Frequenstist vs Bayesian upper limit - **Exercise**
26/03 27/03	- Bayesian priors and posterior construction - Profile likelihood and Wilks' theorem - Bayesian parameter estimation	- Bayesian model selection and evidence - Marginal likelihood, Bayes factor - **Exercise**

Hands-on approach

- Statistics can feel like a cook-book
  - Plenty of ready-to-use recipes - all valid if you understand assumptions, limitations and the question they answer
  - No lengthy demonstrations or one-pager calculations in this course
  - Learn to understand and test different statistical approaches
- Learn by doing = hands-on exercises with python in Jupyter notebooks
- Requirements: Laptop, Unimib account, access to WiFi, *Google Colab*





Note on the use of AI

- Large Language Models (AI)
  - Very powerful today and improving quickly
  - Important tool for learning (but must be very careful)
  - Must be integrated in modern analysis workflow
- **AI as a helpful coding pal (not an oracle)**
  - Great at drafting and explaining code
  - Useful for code autocomplete and refactoring
  - A very confident programmer, will guess incorrectly
  - Often gives wrong answers that look correct
  - It will mirror your level (shallow question → shallow answer)
  - It cannot replace understanding concepts



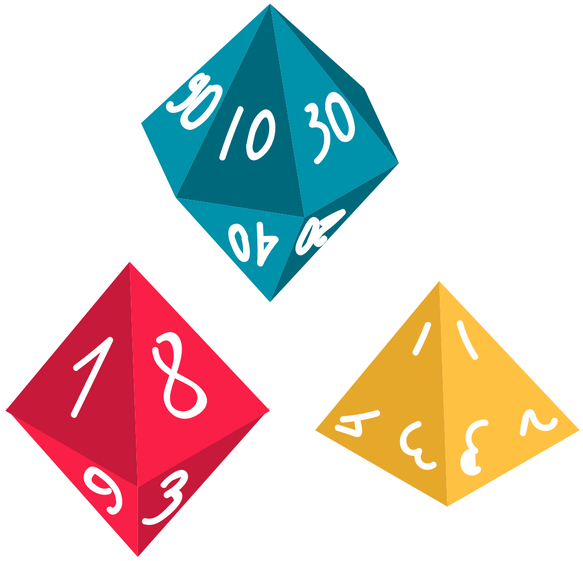
Probability

Foundations and interpretations

Main reference: G.Cowan chapter 1

Why Probability?

- Incorporating **uncertainty** in statistical analysis → **Probability**
- Measurements not exactly reproducible:
  - Fundamental randomness of QM
  - Classical random effects (apparatus)
- Theory prediction uncertainty:
  - Approximations (e.g. perturbation theory)
  - Uncertain response of measurement apparatus



recap

Axiomatic definition of probability

Axioms defined by Kolmogorov in 1933 within the context of *set theory*

- Consider a set S (sample space) and subsets $A, B \subset S$

1. For all subsets we assign a probability $P(A) \geq 0$
2. The probability of the sample space $P(S) = 1$
3. If $A \cap B = \varnothing$ then $P(A \cup B) = P(A) + P(B)$

- Derived properties:

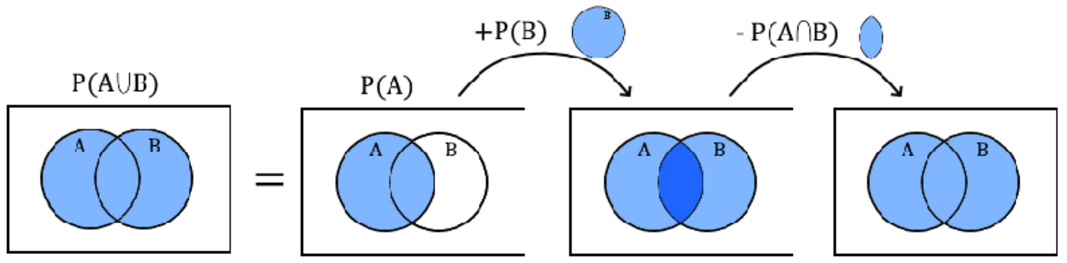
$$P(\bar{A}) = 1 - P(A)$$

$$P(A \cup \bar{A}) = 1$$

$$P(\varnothing) = 0$$

if $A \subset B$, then $P(A) \leq P(B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



recap

Conditional probability

- **Conditional** probability of A given B:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- Example: What is the probability that a card is a queen given that it is a face card?

  {Q}/{J,Q,K} = 4/12 = 1/3

- A and B are **independent** if

$$P(A \cap B) = P(A)P(B)$$

- If A and B independent then:

$$P(A \mid B) = \frac{P(A)P(B)}{P(B)} = P(A)$$

- Requiring B has no effect on the probability of A

recap

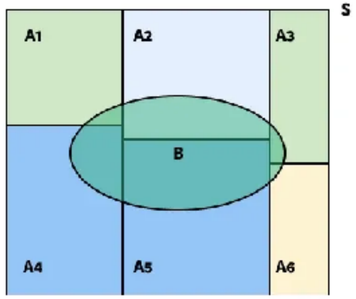


Bayes' theorem (1763)

- Bayes' theorem follows from the definition of conditional probability

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

- **Law of total probability**, assuming S divided into disjoint subsets A_i



$$P(B) = \sum_i P(B \mid A_i)P(A_i)$$

Bayes' theorem becomes:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{\sum_i P(B \mid A_i)P(A_i)}$$

recap

The medical test example

Solve and answer poll on wooclap:
https://app.wooclap.com/NHLRLA



Bayes' theorem:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{\sum_i P(B \mid A_i)P(A_i)}$$

Interpretations of probability

Main reference: G.Cowan chapter 1

Interpretations of probability

- Kolmogorov axioms are abstract
  - No meaning of elements of samples space.
  - No recipe to assign numerical values to probability.
  - Need for an interpretation of probability!

Kolmogorov axioms

- Consider a set S (sample space) and subsets $A, B \subset S$

1. For all subsets we assign a probability $P(A) \geq 0$
2. The probability of the sample space $P(S) = 1$
3. If $A \cap B = \varnothing$ then $P(A \cup B) = P(A) + P(B)$

Frequentist vs Bayesian

- Two important interpretations of probability are used:

1. **Frequentist (classical) interpretation**
   - A, B are outcomes of repeatable experiment

$$P(A) = \lim_{n\to\infty} \frac{\text{times outcome is } A}{n}$$

   - Natural way of thinking in particle physics and in QM
   - Infinite limit is idealistic and in some cases it is meaningless

2. **Bayesian (subjective) interpretation**
   - A, B are statements that are true or false

$P(A) =$ degree of belief that A is true

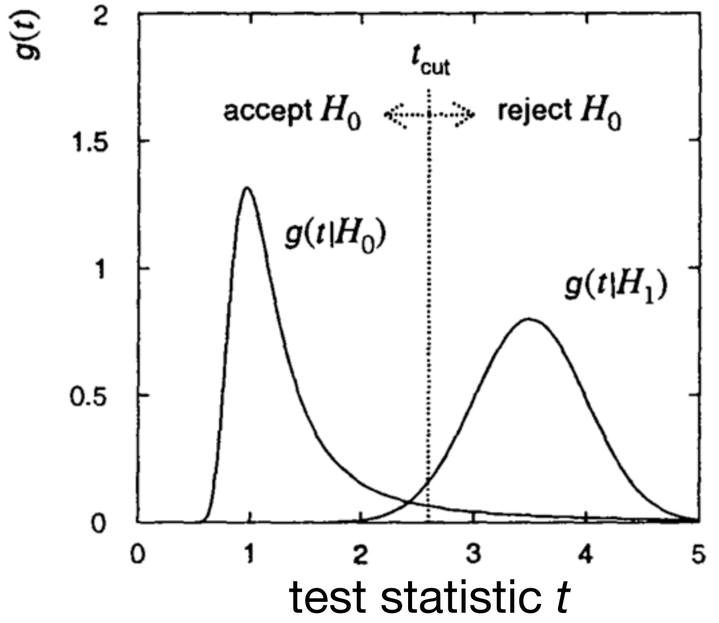
   - Examples: What is the prob that tomorrow it will rain? What is the prob that a particle exists?
   - Probability reflects the degree of belief that A is true

Interpretations of probability lead to different ways of doing stats!

Frequentist philosophy

- **Data** $\vec{x}$ = outcomes of repeatable observations
- **Hypotheses** H = models or parameter values giving certain probabilities for outcomes
- In frequentist statistics probabilities are associated only with the data, not with hypotheses
- Cannot talk about *P(*supersymmetry*),* *P(*9.7<*g*<9.9*), P(*rain tomorrow*)*
- Preferred theories are those that predict a high probability for data “like” the data observed

Frequentist hypothesis testing



Bayesian philosophy

- Bayesian statistics can assign a probability to a hypothesis, but requires a prior (subjective) probability on the hypothesis:

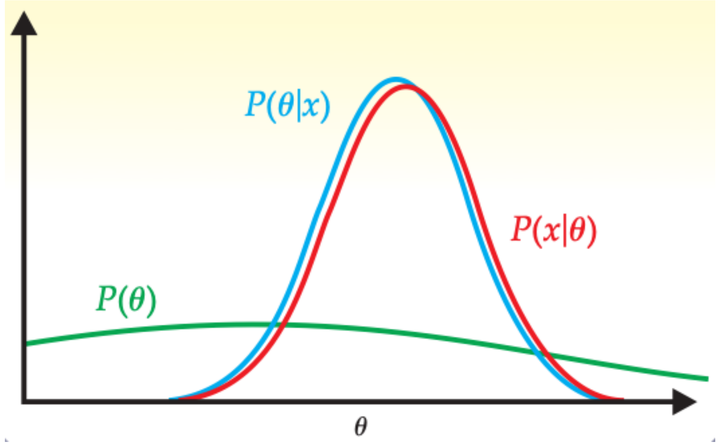
probability of the data assuming hypothesis H (the likelihood)

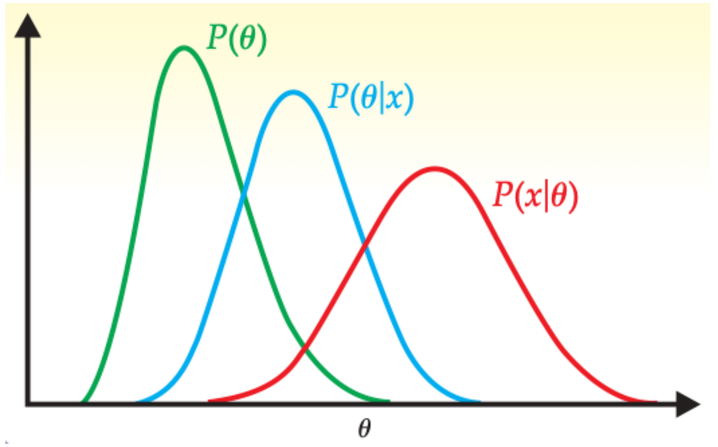
prior probability, i.e., before seeing the data

$$P(H|\vec{x}) = \frac{P(\vec{x}|H)\pi(H)}{\int P(\vec{x}|H)\pi(H)\,dH}$$

posterior probability, i.e., after seeing the data

normalization involves sum over all possible hypotheses





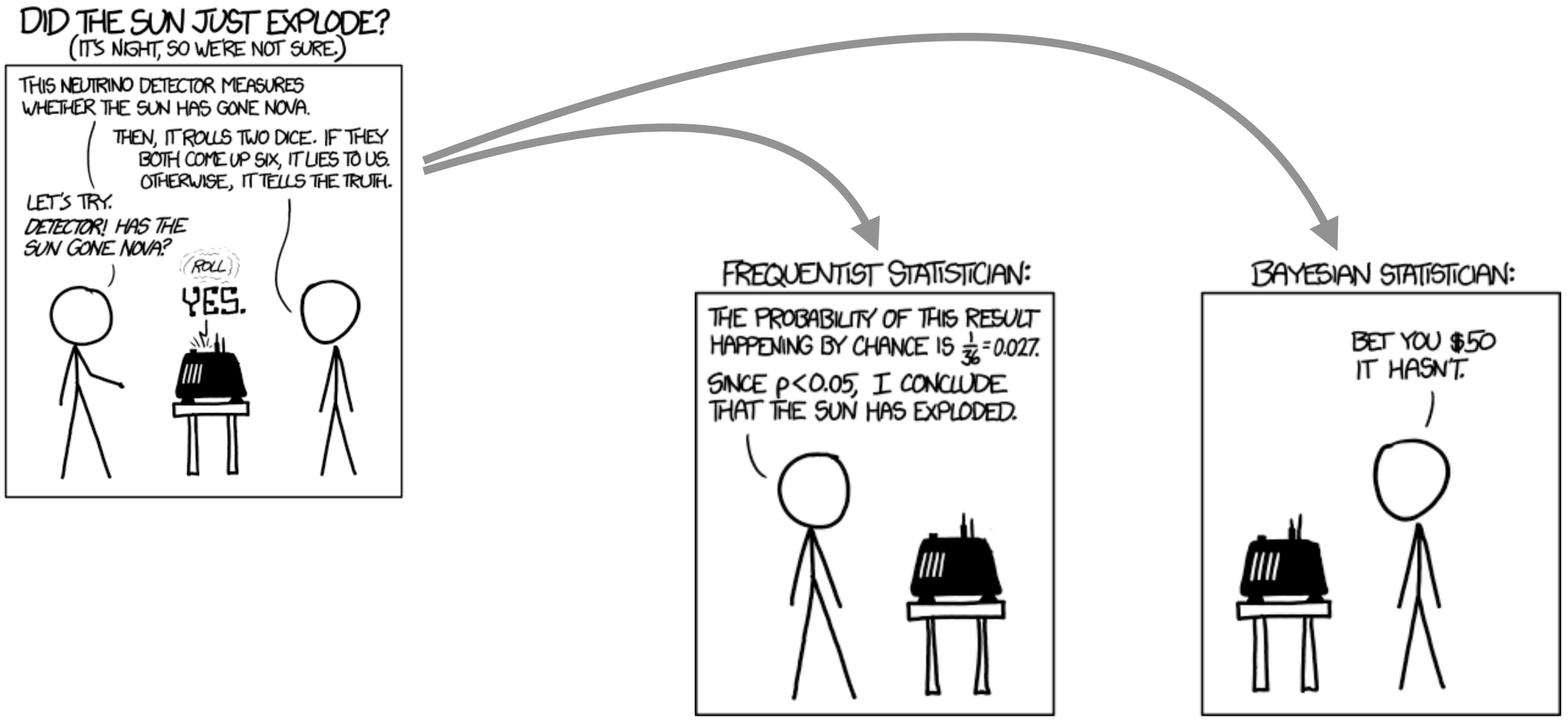
- If you prior probability of H is $\pi(H)$, then in light of the data you can update the probability to $P(H|\vec{x})$
- However: no general prescription for prior probability!

The medical test example



- Bayes theorem is valid regardless of probability interpretation!
- We got that $p(D \mid +) = X\%$
- **Your viewpoint (Bayesian)**:
  my degree of belief that I have the disease is X%
- **Your doctor's viewpoint (Frequentist)**:
  X% of people getting a positive result have the disease

Did the sun just explode?



Frequentist vs Bayesian

- Scientists should not be required to label themselves as frequentists or Bayesians
- There is no right or wrong, the two approaches answer different (but related) questions
- Inference methods are different, but match in the limit of large data samples and weak priors
- The likelihood is the unifying principle





Jupyter notebook

Week1_probability.ipynb

recap

MC methods

Main reference: G.Cowan chapter 3

recap

The principle of MC sampling

MC sampling is a computational method to estimate probabilities using a large sample of random numbers

MC sampling typically consists of these steps:

1. Generate a sequence of *independent* random numbers $r_1, \ldots, r_n$ uniform in $[0,1]$
2. Use them to produce another sequence $x_1, \ldots, x_n$ distributed according to the PDF $f(x)$ we are interested in.
3. Use the $x_1, \ldots, x_n$ sequence to estimate some property of $f(x)$

For example: $\int_a^b f(x)dx$ will be given by the fraction of x values found in the interval $[a, b]$.

Formally:

MC calculation = numerical integration

In practice we will often use MC generated values to calculate properties of complicated PDFs and also as **simulated data** to test our statistical procedures

recap

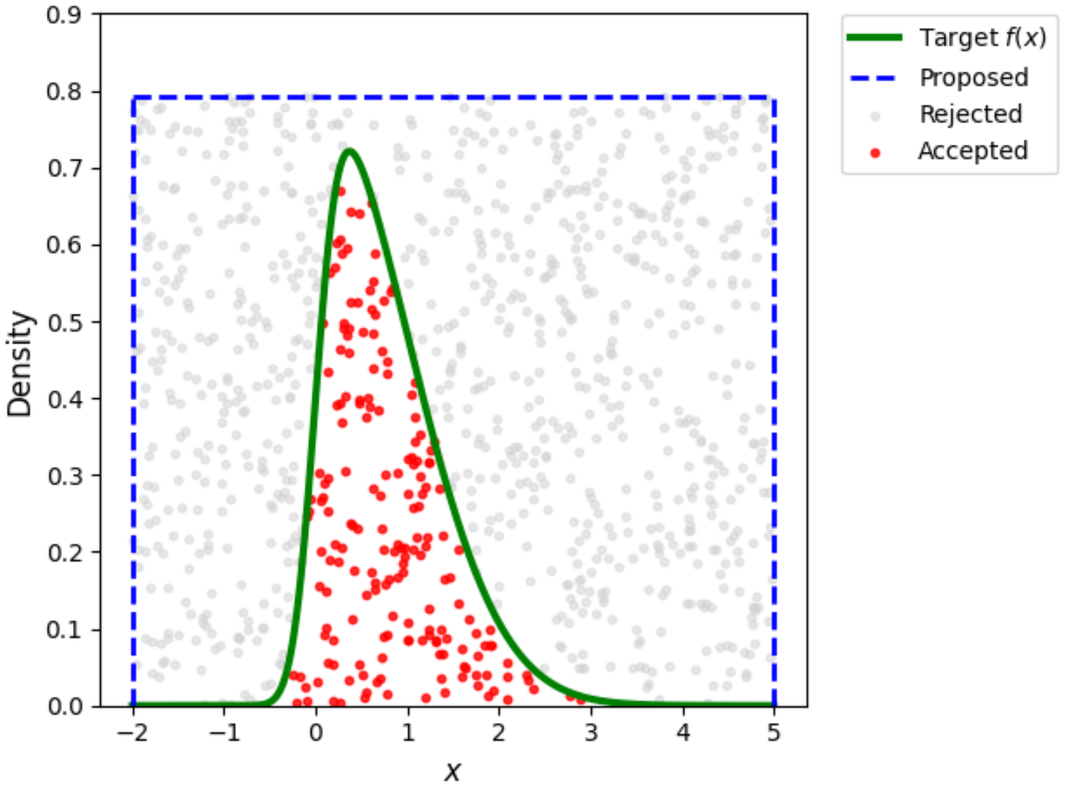
MC sampling
Step 1: generating random numbers

- Computers use pseudorandom number generators (PRNGs): deterministic algorithms designed to mimic true randomness.
- PRNGs start from a seed; the same seed produces the same sequence (important for reproducibility).
- Modern PRNGs have long periods and excellent statistical properties (uniformity, minimal correlations).
  - Examples:
    - **NumPy** (`numpy.random`, default: PCG64),
    - **ROOT** (`TRandom3`, based on Mersenne Twister).



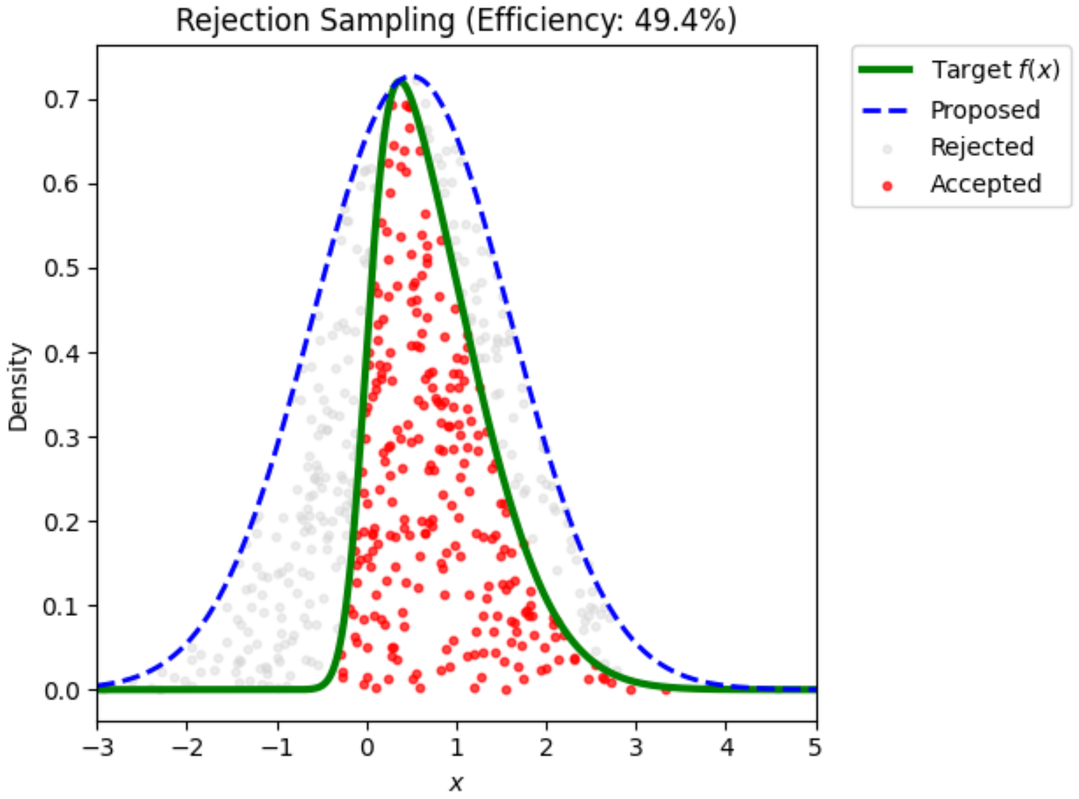
MC sampling
Step 2: produce random variates from PDF

- **The hit-or-miss method** recap
  1. Generate uniform random number x_i in range that covers the domain of the target $f(x)$
  2. Find M factor so that $M \geq \max f(x)$
  3. For each x_i generate uniform y_i in range between 0 and M
  4. Keep x_i only if $y_i < f(x_i)$



MC sampling
Step 2: produce random variates from PDF

- **The rejection sampling method**
  1. Generate random number x_i according to proposed PDF $g(x)$ that is similar to target $f(x)$
  2. Find M factor so that $M \cdot g(x) \geq f(x)$ for any x
  3. For each x_i generate uniform y_i in range between 0 and $M \cdot g(x)$
  4. Keep x_i only if $y_i < f(x_i)$



recap

MC sampling

Step 2: produce random variates from PDF

- **The transformation method**

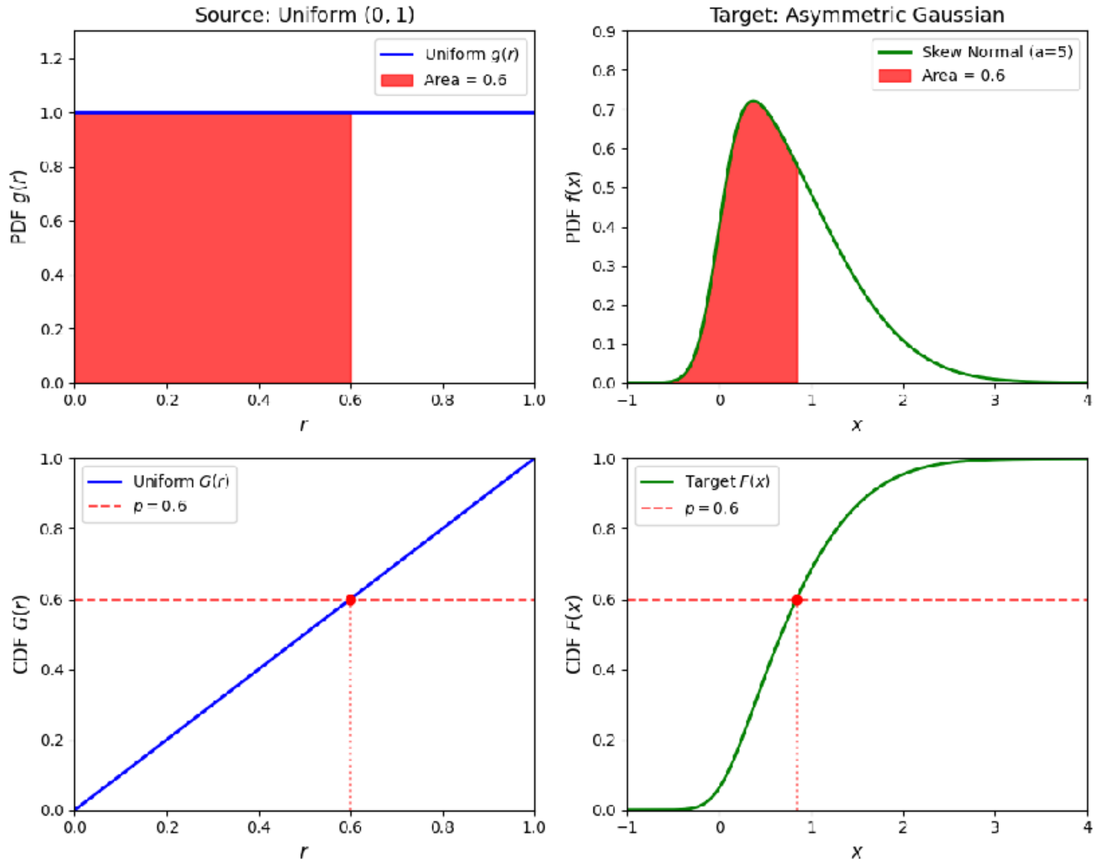
Find the transformation $x(r)$ that transforms the r_i that follow a uniform distribution $g(r)$ in x_i that follow the target PDF $f(x)$

→ **Solution**: match the CDFs and solve for $x(r)$

$$\int_{-\infty}^{r'} g(r)dr = \int_{-\infty}^{x(r')} f(x')\,dx'$$

$$r = F(x(r))$$

→ We "just" need to integrate $f(x)$ to get its CDF $F(x)$ and then invert it to get $x(r) = F^{-1}(r)$



recap

MC sampling

Step 3: use sample to estimate PDF properties

- Estimating statistical properties such as:
  - Expected value $E[x] \simeq \frac{1}{n}\sum x_i$
  - The probability that x falls in interval $[a, b]$ is simply the fraction of samples in that interval
- Propagate uncertainties through nonlinear transformations
- Estimate PDF of derived quantities
  - Estimate PDFs of complicated functions of random variables
  - Incorporate random effects due to measurement uncertainty
- Simulate realistic datasets
  - Simulate repeated experiments
  - Calculate acceptance, resolution, …
- Validate statistical inference method
  - Quantify expected sensitivity of the measurement
  - Check properties of esitmators (e.g. bias)
  - Get distributions of test statistics for hypothesis testing
- Bayesian computation
  - Compute posteriors, credible intervals, …
  - Markov chain MC
- Use for training Machine Learning algorithms…



Jupyter notebook

Week1_MCmethods.ipynb

Exercise for week 1

Week1_exercise.ipynb

